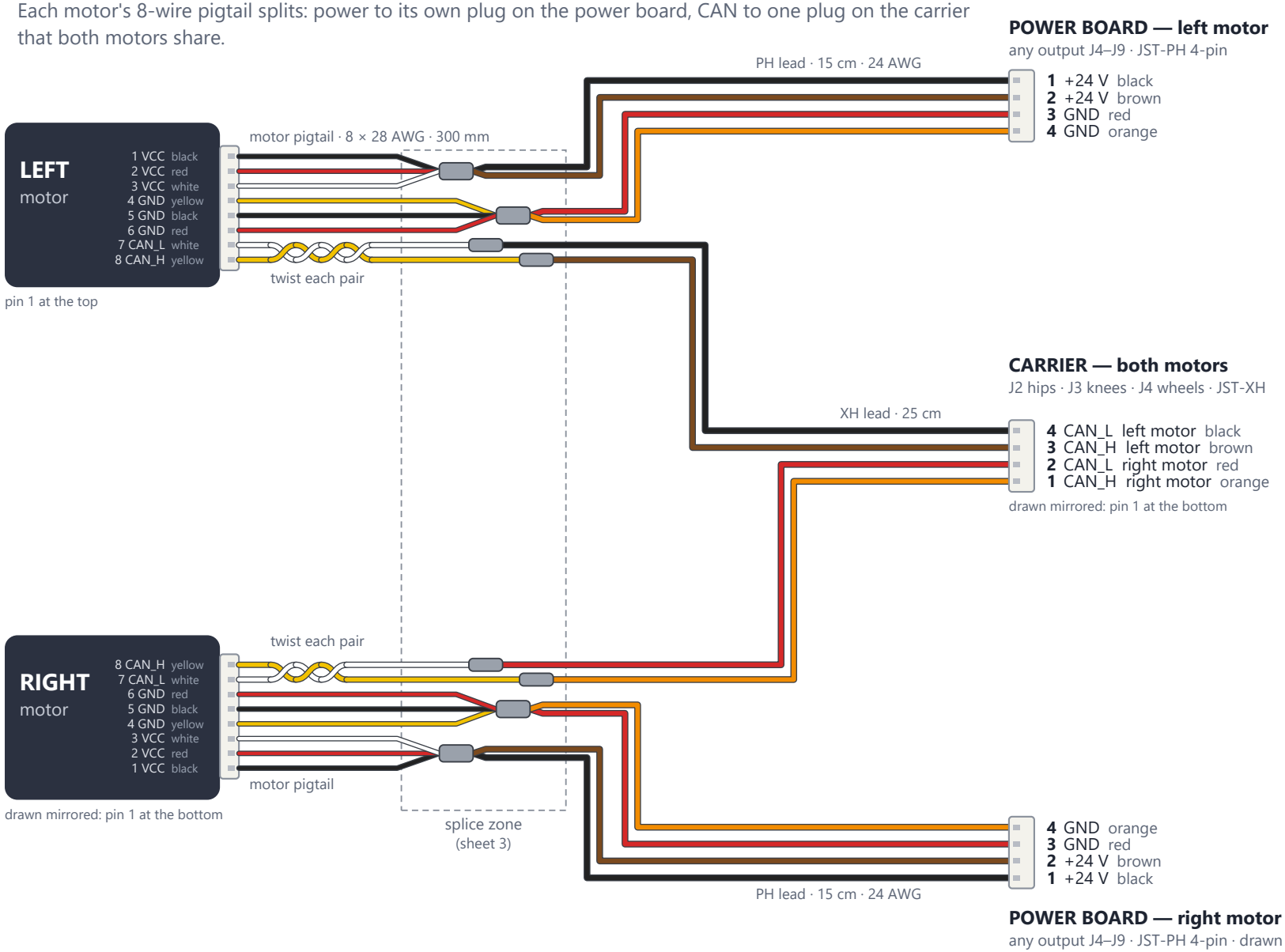
Motor-pair harness — what goes where

Each motor's 8-wire pigtail splits: power to its own plug on the power board, CAN to one plug on the carrier that both motors share.



Five things that must be right

1. **Tag before you cut.** Every motor-lead colour appears twice, in two different jobs, and on the power lead red is ground. Tag all eight motor wires with their pin numbers first.
2. **Power: three into two.** Motor black, red and white (pins 1–3) join the power lead's black and brown. Motor yellow, black and red (pins 4–6) join its red and orange.
3. **CAN: yellow to H, white to L.** Yellow joins the CAN lead's brown (left motor) or orange (right); white joins black or red. On this lead black is pin 4, not pin 1. A swapped pair reads the same on a meter, and that motor never answers.
4. **Both terminators on.** Each motor keeps its 120 Ω switch ON. Nothing on the carrier terminates the bus.
5. **Two different CAN IDs.** Out of the box both answer at ID 1. Renumber one on the bench before they share a bus.

Every wire is drawn in its real colour, in the orientation confirmed on 2026-10-22 and, for the CAN lead, corrected on 2026-10-23. Each pin's signal and colour are printed beside it.

solder joint under heat-shrink

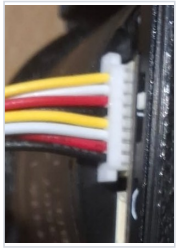
Sources: motor manual V1.0 p. 4 (8-pin order) · docs/POWER_SWITCH.md (PH outputs) · docs/CARRIER.md (J2–J4) · lead colours from the owner's photos

Which of your wires is which

Orientation confirmed on 2026-10-22: on the motor pigtail the CAN pair is at the yellow edge, and on the power lead the black wire is on pin 1. **The CAN lead is the other way round**, orange on pin 1 and black on pin 4: corrected on 2026-10-23, after a lead taken for black-on-1 left both motors deaf.

A Motor pigtail

JST-SH 8-pin, 1.0 mm pitch · plugs into the motor



your photo

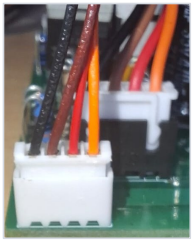


Pin	Signal	Wire	Joins
1	**VCC**	black	+24 V joint, power lead · black 1 · brown 2
2	**VCC**	red	+24 V joint, power lead · black 1 · brown 2
3	**VCC**	white	+24 V joint, power lead · black 1 · brown 2
4	**GND**	yellow	ground joint, power lead · red 3 · orange 4
5	**GND**	black	ground joint, power lead · red 3 · orange 4
6	**GND**	red	ground joint, power lead · red 3 · orange 4
7	**CAN_L**	white	CAN lead · black 4 left · red 2 right
8	**CAN_H**	yellow	CAN lead · brown 3 left · orange 1 right

Every colour appears twice, in two different jobs. Black and red are each +24 V once and ground once; white is +24 V and CAN_L; yellow is ground and CAN_H. Tag all eight wires with their pin numbers before separating any of them.

B Power lead

JST-PH 4-pin, 2.0 mm · one per motor · plugs into the power board



your photo



Pin	Signal	Wire
1	**+24 V**	black
2	**+24 V**	brown
3	**GND**	red
4	**GND**	orange

Red is ground on this lead. Black and brown are +24 V; red and orange are ground.

C CAN lead

JST-XH 4-pin, 2.5 mm · one for both motors · plugs into the carrier



your photo



Pin	Signal	Wire
1	**CAN_H right**	orange
2	**CAN_L right**	red
3	**CAN_H left**	brown
4	**CAN_L left**	black

Orange and brown are CAN_H; red and black are CAN_L. Black is pin 4, at the end away from the carrier's white dot. The carrier joins pin 1 to 3 and 2 to 4, so either pair serves either motor: here the left has 3 and 4.

Finding pin 1. On both boards it is the squared-off pad on the underside; the other pads are oval. On the carrier there is also a white dot printed beside pin 1 of each connector. These tables hold only for this orientation: a lead found the other way round is re-checked, not wired from them.

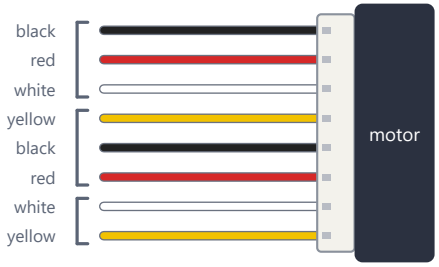
Sources: motor manual V1.0 p. 4 (8-pin order) · docs/POWER_SWITCH.md (PH outputs) · docs/CARRIER.md (J2–J4) · lead colours from the owner's photos

Confirm each lead, then join them

Battery unplugged and nothing powered for all of this. Meter on continuity (beep) unless it says ohms.

1 Confirm each lead before you cut

A beep test takes a minute a lead and catches one made the other way round. **If a lead does not beep like this, stop.** Plug the pigtail into its motor and beep each wire against its neighbours.



black, red and white beep together: +24 V, pins 1–3

yellow, black and red beep together: ground, pins 4–6

white and yellow beep to nothing else, with about 120 Ω between them: CAN_L (white, 7) and CAN_H (yellow, 8)

A short chirp between +24 V and ground is the motor's capacitors charging; a steady beep is a fault. Thousands of ohms between white and yellow instead of about 120 means that motor's termination switch is off: switch it on.

Each power lead, plugged into a motor output on the power board: red and orange beep to the negative of the board's battery input; black and brown do not.

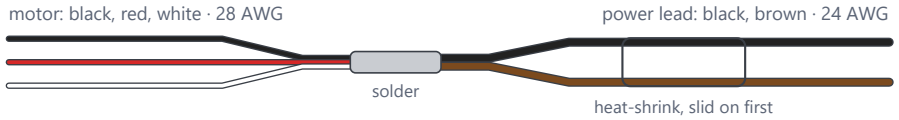
The CAN lead, plugged into the carrier: black beeps to red and brown to orange, because the board joins pin 1 to 3 and 2 to 4. **Orange sits on the pin with the white dot**, black at the far end. The beeps are the same whichever end black is at, so the dot is the only check of it.

2 The joints, wire by wire

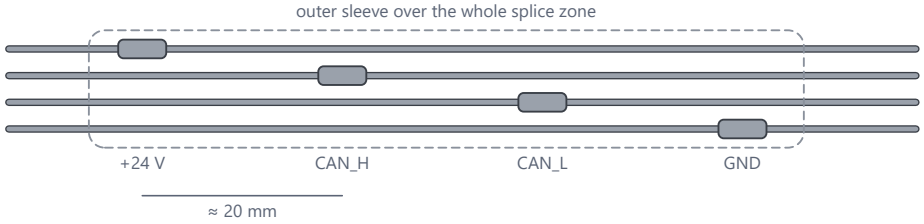
Joint	Motor pigtail wires (pin)	Lead wires (pin)
+24 V joint	black 1 · red 2 · white 3	power lead · black 1 · brown 2
Ground joint	yellow 4 · black 5 · red 6	power lead · red 3 · orange 4
CAN_H	yellow 8	CAN lead · brown 3 left · orange 1 right
CAN_L	white 7	CAN lead · black 4 left · red 2 right

Each motor has its own power lead; the one CAN lead takes both motors, the left on pins 3–4 and the right on 1–2.

Making the joints



The +24 V joint: twist the three motor wires together, twist the two lead wires together, overlap them in line about 8 mm, solder, then shrink. Ground is made the same way.



- **Stagger the joints** about 20 mm apart so the bundle never doubles up in one place, with the two CAN joints between the +24 V and ground joints so the two power joints never lie side by side.
- **Slide every piece of heat-shrink on before soldering**: one per joint, and one larger sleeve per motor over the whole zone.
- **Twist each motor's yellow with its white**, about a turn every 2–3 cm, and keep twisting through the joints with the CAN lead's matching pair.
- **Lay the cable along its real route before cutting.** The 300 mm pigtail is expected to reach the torso from a hip or a knee but not from a wheel, and none has been laid along a leg yet. Leave slack where it crosses a joint.
- **Keep strain off the motor plug**: its contacts are crimped onto 28 AWG. Tie the pigtail down near the motor with a little slack.
- **Label each end**: left power, right power, CAN, and which pair of joints it is for.

Sources: motor manual V1.0 p. 4 (8-pin order) · docs/POWER_SWITCH.md (PH outputs) · docs/CARRIER.md (J2–J4) · lead colours from the owner's photos

Check it before it sees power

Both motors plugged in, nothing powered, and the cable not yet plugged into either board.

Measure at	Between	Expect	If not
CAN plug	pins 3 and 4: brown, black	about 120 Ω: the left motor's terminator	Open: a CAN joint is broken, or that motor's switch is off. Near 0: H and L are shorted.
CAN plug	pins 1 and 2: orange, red	about 120 Ω: the right motor's terminator	As above, for the right motor.
Each power plug	pins 1 and 2: black, brown	beep	A +24 V joint is open.
Each power plug	pins 3 and 4: red, orange	beep	A ground joint is open.
Each power plug	black against red	no steady beep; a short chirp is the motor's capacitors	**+24 V is shorted to ground. Do not power it.**
Power plug to CAN plug	every wire against every wire	no beep	**Power is on a CAN wire. Do not power it:** this is the fault that would destroy the transceivers.

Then plug the CAN lead into the carrier (still unpowered) and measure across the back of the plug, black to brown: **about 60 Ω**, the two terminators in parallel. 120 Ω means one motor's pair is open or its switch is off.

> **No meter check finds a swapped CAN_H and CAN_L**: the pair reads the same 120 Ω either way round. Only joining by the list on sheet 3 prevents it, and only a motor answering over the bus proves it. On 2026-10-23 both motors were joined from a CAN lead read the wrong way round, and every check on this sheet passed.

Before the two motors share a bus

- **Give them different CAN IDs**, one motor at a time on the bench with the USB-to-CAN debugger: the motor GUI's 'Give this motor a new CAN ID', which sets its feedback ID at the same time. Out of the box both answer at ID 1, and two motors at one ID on one bus collide. Left 1, right 2 is a suggestion; no ID plan is recorded yet.
- **Check both termination switches are ON.** The first table proves it: 120 Ω on each pair.

First power-up

1. Robot off. Plug in the motor ends, then the two power plugs, then the CAN plug.
2. Switch on with the button. Never plug or unplug a motor while the bus is on.
3. Both motors should answer. On the wheel bus the board scans for them itself (sbc/wheel_check.py --scan). Elsewhere, prove the cable with the bench USB-to-CAN debugger in the carrier's place: its CAN_H on brown and CAN_L on black for the left motor, then orange and red for the right, and scan with the motor GUI.
4. If one motor is silent and the other answers, suspect that motor's CAN pair before anything else.

Sources: motor manual V1.0 p. 4 (8-pin order) · docs/POWER_SWITCH.md (PH outputs) · docs/CARRIER.md (J2–J4) · lead colours from the owner's photos